Raft Consensus Algorithm

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- Subject: Distributed systems
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What Problem Does Raft Solve?

Distributed systems have multiple nodes

Nodes can fail or lose connection

Data can become inconsistent

What is Distributed Consensus?

Process where multiple nodes agree on:

- same data
- same order of operations

Without consensus:

- Data conflicts
- System inconsistency

With consensus:

- Reliable and consistent system

Raft Roles

Three types of nodes:

Leader

- Handles requests
- Controls system

Follower

- Passive, follows leader

Candidate

- Competes to become leader

Leader Election

- Followers wait for leader heartbeat
- If no heartbeat → become candidate
- Candidate requests votes
- Majority votes → becomes leader

Log Replication

- Leader receives client request
- Adds it to its log
- Sends to followers
- Followers copy the log

Committed Entries

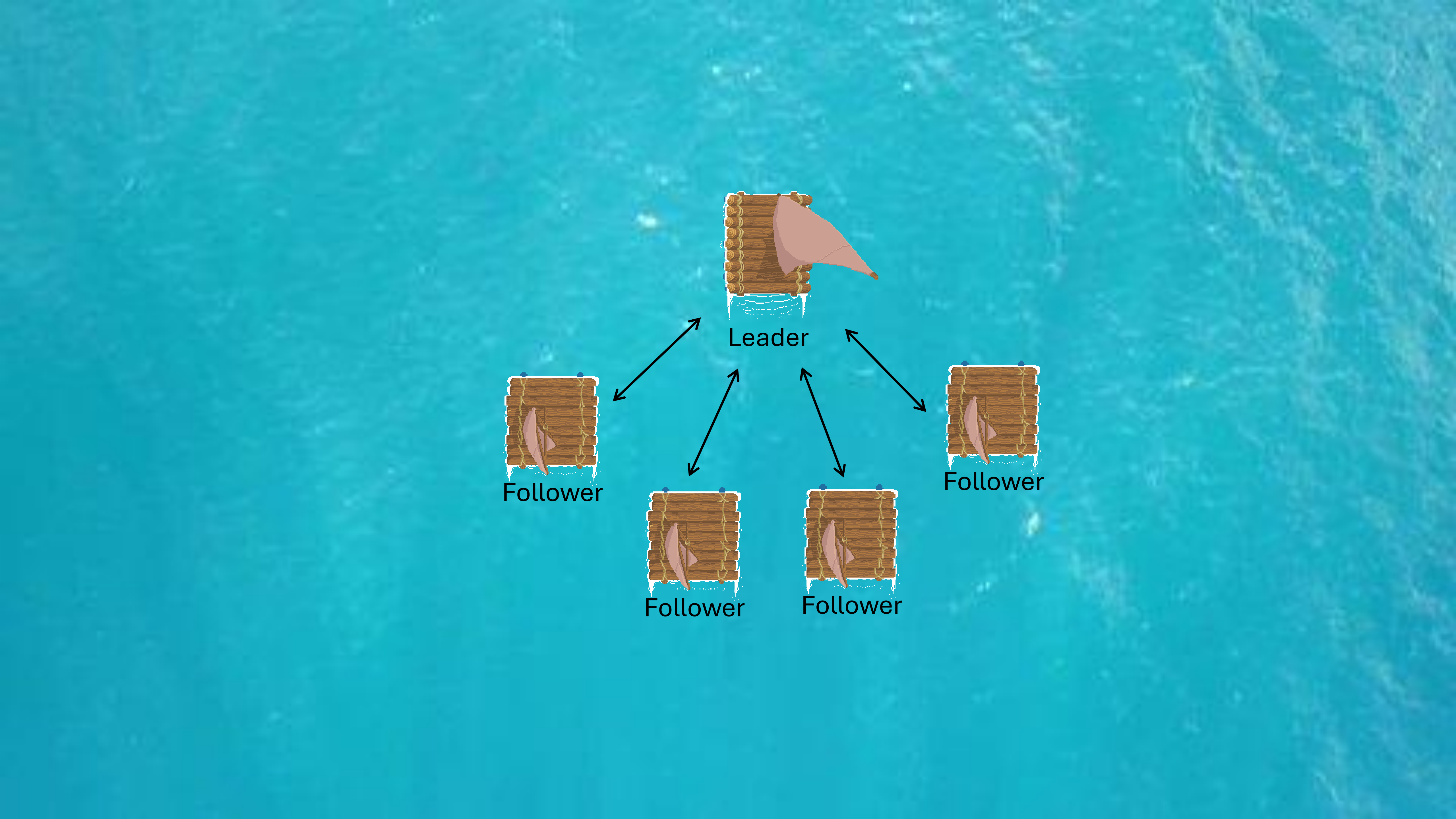
Entry is committed when:

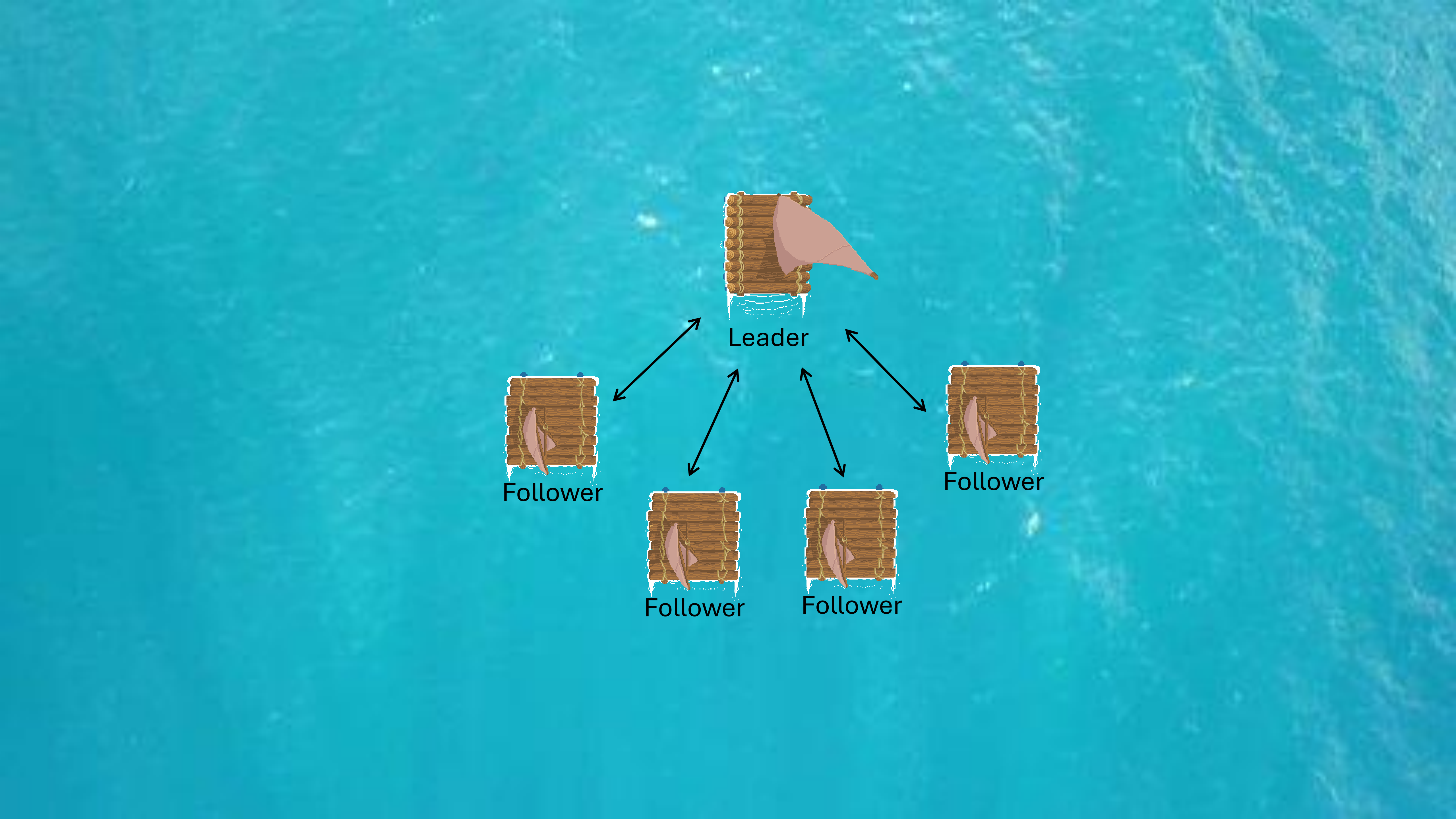
stored on **majority of nodes**

After commit:

Safe to apply changes

Data becomes consistent


Leader
Follower
Follower
Follower
Follower



Leader
Follower
Follower
Follower
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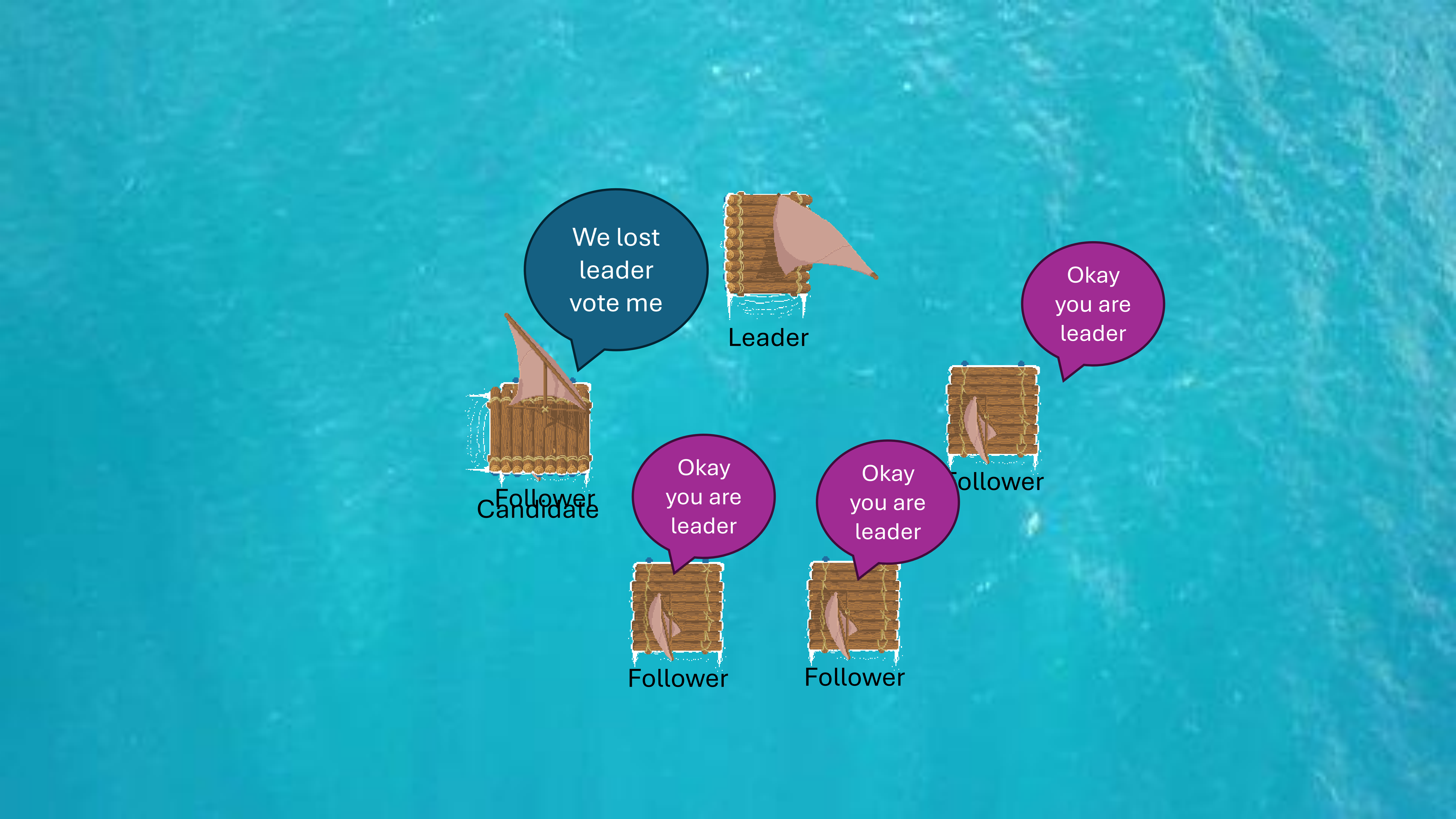
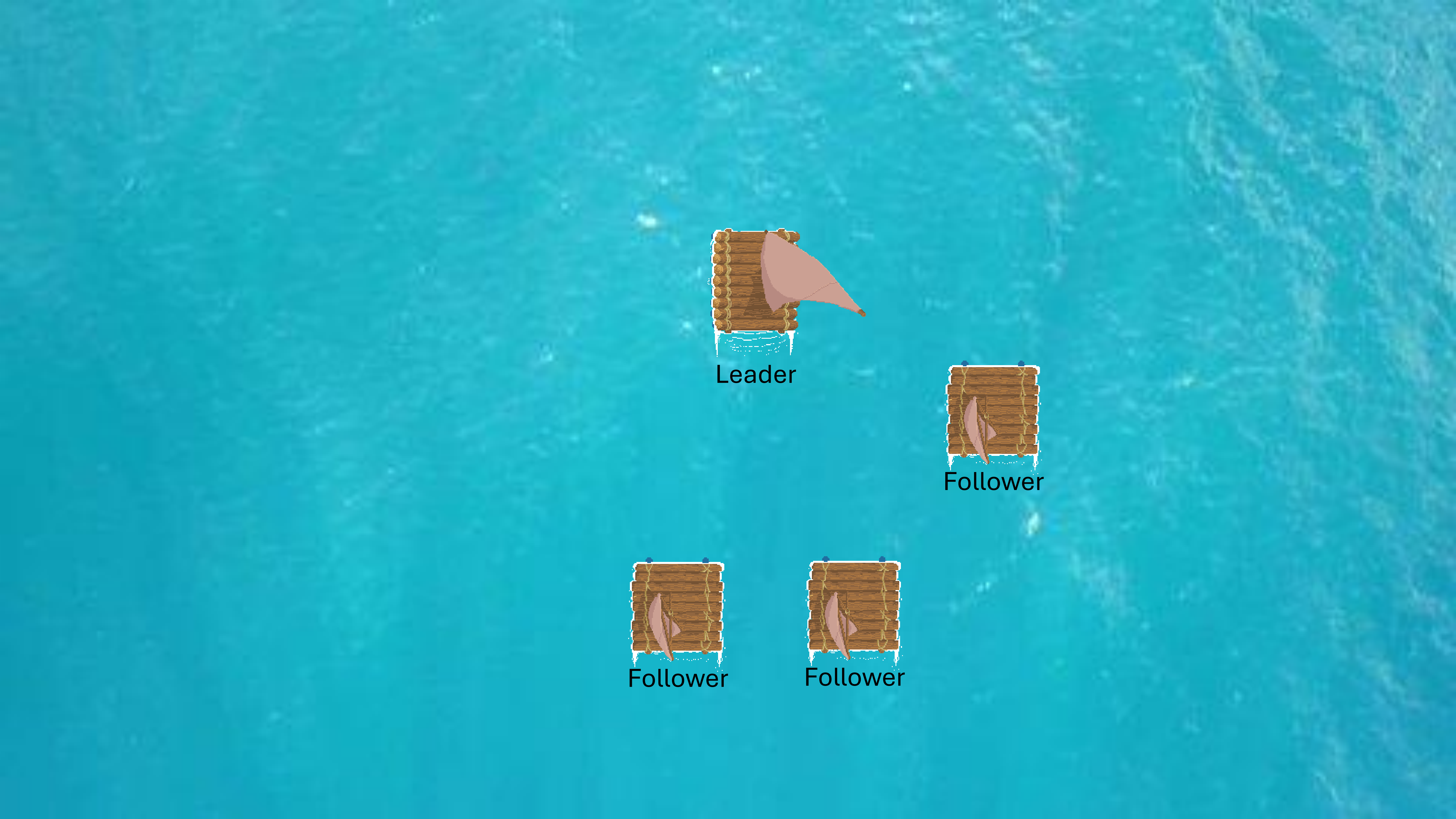

We lost leader vote me
Leader
Follower
Candidate
Okay you are leader
Okay you are leader
Okay you are leader
ollower
Follower
Follower


Candidate
Follower
Follower
Follower


Leader
Follower
Follower
Follower